

Ultra Gym: Transforming Clinical 3D Ultrasound Volumes into Dynamic, Multi-Task Training Data

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Abstract. Three-dimensional (3D) ultrasound (US) volumes offer rich, comprehensive anatomical information, yet they remain heavily underutilized in medical AI development. This underutilization stems from a lack of dedicated data-engineering frameworks capable of converting static, archived volumes into interactive, multi-task training data. Existing methods commonly rely on cross-modality synthesis of ultrasound images from CT or MRI data. This practice introduces simulation-to-real domain gaps and fails to preserve native ultrasound-specific acoustic artifacts. Furthermore, alternative approaches are generally restricted to isolated, single-task applications. To address these data-engineering limitations, we present Ultra GYM, a framework that transforms static 3D ultrasound archives into dynamic, interactive training environments. By operating directly on clinical 3D volumes, our platform preserves native acoustic properties and eliminates cross-modality domain shifts. Ultra GYM provides a modular data-engineering framework designed to easily extend to broader medical imaging pipelines. To showcase this multi-task scaling capability, we configure the framework to automatically generate specialized data structures for three highly distinct downstream tasks: (1) temporal video sweeps for standard plane detection, (2) exact six-degree-of-freedom tracking coordinates for 2D to 3D US reconstruction, and (3) an efficient reinforcement learning environment for autonomous Standard Plane detection. Across downstream evaluations, Models trained via the framework achieved 79.47% overall accuracy in standard plane detection, an 84% success rate in autonomous localization, and a local pose reconstruction error of about 0.12 mm. These results demonstrate that Ultra GYM serves as an effective integrated framework, extracting diverse, multi-task training trajectories from limited clinical archives to accelerate medical AI deployment. The dataset, code, and videos will be made publicly available upon publication.

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Keywords: Ultrasound Simulation Environment · Standard Plane Detection · Autonomous Ultrasound Navigation · Ultrasound Reconstruction · Reinforcement Learning

1 Introduction

Three-dimensional ultrasound volumes (US) are increasingly common in clinical workflows, providing rich spatial views of internal anatomy [7,8]. However, these static archives remain heavily underutilized for developing computational methods or training advanced machine learning models [2]. Downstream medical AI tasks—such as standard plane detection, 3D reconstruction, and autonomous standard plane tracking—depend heavily on dynamic, task-specific data. Crucially, each requires a distinct data architecture: standard plane detection demands continuous video streams with precise temporal labels; 3D reconstruction requires dense frames paired with exact spatial tracking coordinates; and autonomous navigation necessitates an interactive environment capable of real-time feedback. Furthermore, prospective acquisition and manual annotation of these specialized datasets is expensive, time-consuming, and operator-dependent [4], presenting a major bottleneck for scalable ultrasound algorithm development.

To bypass the data bottleneck, current simulation practices synthesize 2D ultrasound images from cross-modalities such as CT or MRI [3,16], frequently leveraging generative networks to map from non-ultrasound data arrays [11]. However, converting these external modalities introduces a simulation-to-real domain gap [1]. Synthetic images generated via cross-modality mapping fail to replicate complex, native acoustic physics, such as direction-dependent speckle noise, acoustic shadowing [14], and signal attenuation, causing downstream models trained on them to generalize poorly to real clinical data [15,17]. Concurrently, learning-based freehand 2D to 3D ultrasound reconstruction, such as TUSRec [9], demands dense frame streams paired with accurate, synchronized probe-pose supervision for volumetric recovery. In practice, this spatial supervision must be captured using external IMU or optical tracking systems [12], which heavily increases acquisition complexity and restricts data scalability.

To address these limitations, we present Ultra GYM, a framework that transforms static 3D ultrasound archives into dynamic, interactive training environments. By operating directly on raw clinical volumes, Ultra GYM inherits native acoustic textures and tissue boundaries, eliminating the cross-modality domain shift typical of CT- or MRI-based simulation. Furthermore, we overlay a dynamic noise mask onto each 2D image to simulate live frame-to-frame acoustic flickering, preventing the feed from appearing artificially static when the probe freezes. By slicing through these noise-augmented volumes, Ultra GYM serves as an extensible framework supporting diverse data-engineering workflows; while designed to scale seamlessly beyond predefined tasks, we demonstrate its multi-task utility here across three distinct downstream pipelines:

- **Continuous Temporal Streams:** We generate automated 2D video sweeps paired with frame-level temporal labels, providing the sequential data necessary to train standard plane detection models.
- **Geometric Spatial Coordinates:** We establish ground-truth six-degree-of-freedom (6-DoF) spatial tracking data paired with corresponding 2D image slice, removing the need for physical tracking hardware for training free-hand 2D to 3D ultrasound reconstruction algorithms.
- **Interactive State Environments:** We build a reinforcement learning Gym directly on the 3D volume, providing real-time feedback for autonomous standard plane localization training.

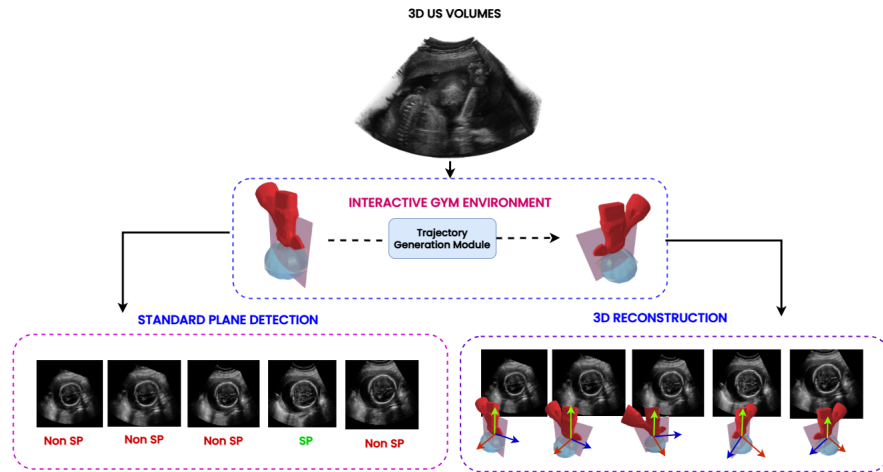


Fig. 1. ULTRA-GYM pipeline overview. A 3D ultrasound volume undergoes volumetric slice rendering, anatomically constrained $SE(3)$ probe motion, and dense trajectory annotation, supporting downstream tasks including standard plane detection and 3D reconstruction.

Across downstream evaluations, Ultra GYM demonstrates strong utility as an integrated data-engineering framework. Models trained via the framework achieved 79.47% overall accuracy in standard plane detection, an 84% success rate in autonomous localization, and a local pose reconstruction error of 0.12 mm. These outcomes prove that Ultra GYM successfully amplifies limited clinical archives into diverse training data structures to accelerate multi-task medical AI deployment.

2 Methodology

2.1 Framework Overview and Gym Environment Design

The core architecture of Ultra GYM maps a static 3D ultrasound volume into a continuous, interactive coordinate space. The environment defines a state S at

any given time step as a combination of a 6-Degree-of-Freedom (6-DoF) Cartesian pose and its corresponding resampled 2D B-mode image slice:

$$S = (\mathbf{x}, I) \quad (1)$$

where $\mathbf{x} = [x, y, z, \alpha, \beta, \gamma]^T$ represents the spatial position and orientation of the simulated ultrasound probe relative to the volume’s origin, and I is the 2D US image extracted for that Cartesian pose.

To allow the agents to navigate this space, the environment handles movement through an action vector $\mathbf{a}$, which applies small spatial translations $(\Delta x, \Delta y, \Delta z)$ and angular rotations $(\Delta\alpha, \Delta\beta, \Delta\gamma)$ to update the probe pose:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{a}_t \quad (2)$$

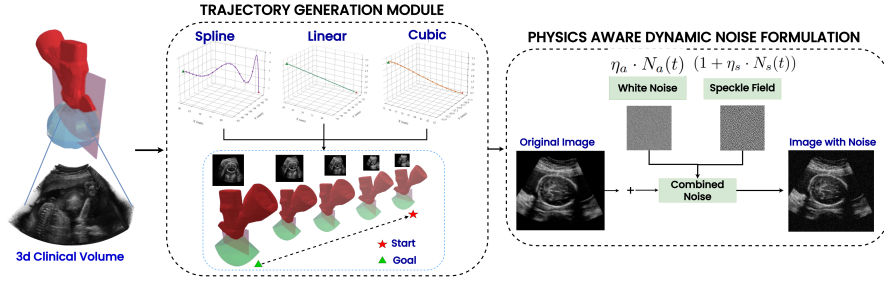


Fig. 2. Pipeline of Ultra GYM: A 3D clinical volume is sampled along simulated probe trajectories to generate ultrasound frames, which are subsequently augmented using a physics-aware dynamic noise model to reproduce realistic temporal ultrasound appearance and speckle dynamics

2.2 Physics-Aware Dynamic Noise Formulation

In live examinations, real ultrasound feeds exhibit frame-to-frame pixel variance driven by electronic thermal noise and tissue micro-motions (e.g., blood flow or respiratory tremors). Standard volumetric data pipelines fail to capture this behavior; slicing through a static 3D clinical volume yields identical pixel arrays whenever simulated probe coordinates remain unchanged. To address this limitation and simulate realistic frame-to-frame temporal variations without complex acoustic wave modeling, Ultra GYM incorporates an empirical dynamic noise model. Rather than computing first-principles wave physics, our approach directly induces two key noise sources observed in live B-mode streams: spatially correlated multiplicative speckle fluctuations and additive electronic noise. For each resampled 2D slice I_t at step t , the noise-augmented frame $I_{\text{final}}(t)$ is generated as:

$$I_{\text{final}}(t) = I_t \odot (1 + \eta_s \cdot N_s(t)) + \eta_a \cdot N_a(t) \quad (3)$$

where $\odot$ denotes element-wise multiplication, $N_s(t)$ is a spatially correlated multiplicative noise field simulating dynamic speckle flickering, and $N_a(t)$ represents

additive electronic noise. The scalars η_s and η_a control the respective noise intensities. As illustrated in Fig. 2, this formulation ensures that even when the probe position is static ($\mathbf{x}_{t+1} = \mathbf{x}_t$), resampled pixel values fluctuate naturally across consecutive steps to match realistic clinical streams.

2.3 Trajectory Generation Module

To collect structured training data across different downstream tasks, Ultra GYM uses an open-ended trajectory generation module. This module connects a starting probe pose $\mathbf{x}_{\text{start}}$ and a target pose $\mathbf{x}_{\text{end}}$ using geometric interpolation functions. As illustrated in Fig. 2, depending on the task requirements, the user can configure the path generation using linear interpolation, cubic splines for smooth operator curves, or parabolic paths to simulate complex anatomical sweeps:

$$\mathbf{x}(t) = f(\mathbf{x}_{\text{start}}, \mathbf{x}_{\text{end}}, t), \quad t \in [0, 1] \quad (4)$$

By sampling along these mathematically defined paths, the framework generates continuous sequences of images and matching coordinate pairs without manual intervention.

2.4 Automated Data Collection for Standard Plane Detection and Segmentation

For standard plane (SP) detection, deep learning models must learn to differentiate diagnostic planes from non-standard frames. Ultra GYM automates this data collection by introducing operator-like variation in a strict sequence. First, the framework initializes the simulated probe at a randomized initial pose at an offset distance from a known target standard plane. Next, the trajectory generation module executes a path starting from this random state that moves toward, passes directly through, and subsequently overshoots the target SP coordinates. Every 2D frame sampled along this path is compiled with frame-level temporal labels indicating its geometric proximity to the true diagnostic target.

Crucially, to mitigate the severe class imbalance caused by transitional frames outnumbering diagnostic standard plane windows, the trajectory generation module samples multiple perturbed views of the standard plane. Executing random micro-trajectories constrained within a tight ϵ -sphere around target SP coordinates synthesizes a diverse subset of clinically valid poses. This targeted data amplification enriches the positive class, enforcing model invariance against minor operator adjustments to improve overall classification reliability.

2.5 Ground-Truth Data Collection for 3D Ultrasound Reconstruction

Freehand 2D to 3D ultrasound reconstruction networks require dense 2D frame sequences paired with highly accurate tracking coordinates to rebuild a 3D volume. In clinical settings, collecting this data requires expensive external spatial

tracking hardware. Ultra GYM bypasses this hardware requirement by working in reverse. The trajectory module executes dense, pre-planned scanning sweeps (such as linear sweeps or fan-like rotations) across the clinical voxels. Because the environment has absolute control over the coordinate system, it outputs a sequence of 2D B-mode frames where every single frame is perfectly paired with its true 6-DoF spatial tracking coordinate $\mathbf{x}_t$:

$$\mathcal{D}_{\text{recon}} = \{(I_1, \mathbf{x}_1), (I_2, \mathbf{x}_2), \dots, (I_M, \mathbf{x}_M)\} \quad (5)$$

2.6 Reward Formulation for the Interactive Gym Environment

In the autonomous standard plane localization task, the agent is tasked with navigating through the volumetric space to locate and reach a specific target standard plane. At each time step, the input provided to the agent consists of a dual-image observation pair: the current resampled image slice I_t and the target goal image I^* . To train this agent within Ultra GYM, we formalize the control problem as a Markov Decision Process (MDP) with a bounded spatial operational zone. We define a maximum search radius $R_{\max}$ and a maximum angular deviation $\theta_{\max}$ around the target standard plane pose $\mathbf{x}^*$. The agent is constrained to operate strictly within these boundaries; crossing them terminates the episode immediately. The total step reward R_t combines positional alignment ($d_t = \|\mathbf{p}_t - \mathbf{p}^*\|_2$), axis-angle orientational difference (θ_t), and an out-of-bounds environment penalty:

$$R_t = R_{\text{pos}} + R_{\text{ori}} + R_{\text{out_env}} \quad (6)$$

where these distinct step-level sub-components are bounded and defined as:

$$R_{\text{pos}} = -\frac{d_t}{R_{\max}}, R_{\text{ori}} = -\frac{\theta_t}{\theta_{\max}}, R_{\text{out_env}} = \begin{cases} -0.5 & \text{if } d_t > R_{\max} \text{ or } \theta_t > \theta_{\max} \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

3 Experiments and Results

Dataset and Clinical Annotations: Our benchmark uses 2nd-trimester 3D fetal head ultrasound volumes collected by expert sonologists at Mediscan, India, led by Dr. Suresh. All reference standard planes were manually annotated and validated by these expert sonologists.

Experimental Setup: The objective of our evaluation is to demonstrate the multi-task utility and data-synthesis capabilities of Ultra GYM across diverse learning paradigms, rather than to introduce or optimize a task-specific model architecture. To validate how effectively the framework repurposes static volumes into distinct, specialized training structures, we evaluate the pipeline’s enablement capacity across three representative applications: (i) temporally labeled video sweeps for standard plane understanding, (ii) image-pose paired trajectories for freehand three-dimensional ultrasound reconstruction, and (iii) interactive environments for autonomous probe navigation.

Temporally Labeled Video Sweep Generation for Standard Plane Detection: To evaluate Ultra GYM’s utility for Standard Plane (SP) detection, we fine-tune a FetalCLIP model [10] using a parameter-efficient adapter [5] in a few-shot setting. Using 20 distinct 3D ultrasound volumes, our framework automatically synthesized two separate dataset configurations for comparative evaluation: (1) an *Unbalanced* dataset containing raw macro-sweeps where valid Standard plane views are severely sparse and heavily outnumbered by transitional frames, and (2) a *Balanced* dataset where Ultra GYM’s trajectory generation module executed randomized micro-movements within a tight ϵ -sphere around target SP coordinates to sample diverse, perturbed diagnostic Standard plane views. For rigorous evaluation, models trained on these configurations were completely validated on an independent, external in-house dataset featuring zero volume overlap with the training archives.

Table 1. Results show substantial gains in accuracy and F1 when micro-trajectories are incorporated.

Data Configuration	Overall Acc.(%)	SP/Non-SP Acc.(%)	F1 (%)
Unbalanced data	43.05	37.36 / 46.15	38.40 $\pm$ 23.40
Balanced data	79.47	83.93 / 71.29	80.20 $\pm$ 9.04

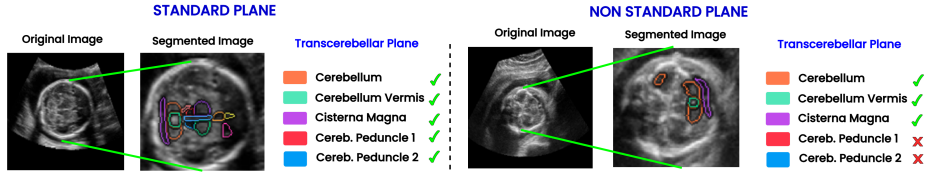


Fig. 3. Anatomical verification of standard and non-standard transcerebellar plane predictions using semantic segmentation. Presence or absence of key cerebellar structures confirms the classification outcome.

As summarized in Table 1, the model trained on the unbalanced baseline struggles significantly under severe class distribution skew, yielding an overall accuracy of just 43.05% and an F1-score of 38.40%. In contrast, when utilizing Ultra GYM’s micro-trajectories to balance the dataset, performance rises substantially to an overall accuracy of 79.47% and an overall F1-score of 80.20%. This absolute F1-score improvement of 41.80% quantitatively demonstrates that targeted spatial perturbation effectively counteracts data sparsity and guides the network to learn robust invariance to minor alignment variations. To verify anatomical validity beyond geometric labeling, predicted SP frames were processed through an in-house segmentation network, confirming the consistent presence of view-specific landmarks (refer Fig. 4).

Ground-Truth Probe Tracking for Freehand 3D Ultrasound Reconstruction: To evaluate Ultra GYM’s utility for freehand 3D ultrasound reconstruction, we trained two trackerless reconstruction baselines, 2D CNN[13] and

Table 2. Trajectory reconstruction performance using trackerless baselines trained on Ultra GYM-generated trajectories. Local PDE measures frame-to-frame reconstruction accuracy, while Final Drift measures accumulated trajectory error.

Method	Local PDE (mm)				Final Drift (mm)			
	Min	Median	Max	Avg.	Min	Median	Max	Avg.
Pairwise CNN	0.03	0.10	0.27	0.14	0.51	7.49	42.00	10.80
DCL-Net Baseline	0.04	0.11	0.22	0.12	0.80	7.04	27.14	8.47

DCL-Net [6], using trajectories generated from Ultra GYM. The models were trained on 1,000 trajectories sampled from 20 clinical volumes and evaluated on 350 trajectories extracted from 5 unseen volumes to assess generalization to unseen patient anatomy. As shown in Table 2, both methods achieve sub-millimetre local reconstruction accuracy (0.14 mm and 0.12 mm average local PDE, respectively), demonstrating that the voxel-native pose annotations provide precise frame-to-frame spatial supervision. Despite accumulated open-loop drift, both baselines maintain average final drift below 11 mm, comparable in magnitude to previously reported sensor-based reconstruction methods. These results indicate that Ultra GYM provides realistic supervision for learning freehand ultrasound reconstruction without requiring external tracking hardware.

Table 3. Autonomous target-plane navigation results within Ultra GYM.

Evaluation Metric	Agent Performance
Navigation Success Rate (%)	84 %
Average Episode Length (Steps)	78 ± 13

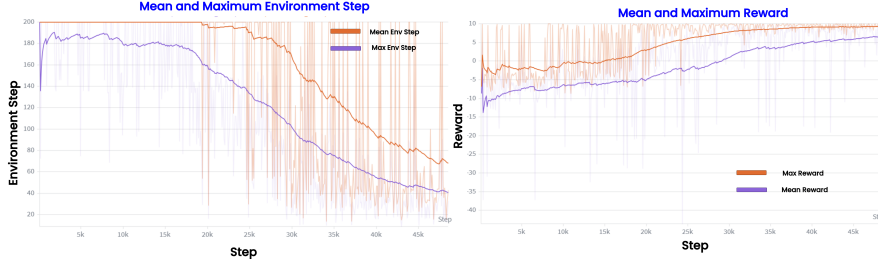


Fig. 4. Ultra GYM navigation training metrics showing rising episodic rewards (right) and decreasing episode step counts (left).

Reinforcement Learning for Autonomous Target Plane Localization:

We evaluate Ultra GYM’s utility as a closed-loop navigation environment by training a deep reinforcement learning policy using a goal-conditioned imitation learning framework with dual-image observations (I_t, I^*) to localize target standard planes. As illustrated by the training curves in Fig. 4, the policy exhibits stable convergence; episodic rewards steadily increase while the mean and maximum environment step counts systematically contract over training iterations. This dual trend confirms that the agent successfully learns to optimize

its physical trajectories, executing progressively shorter paths to reach the goal plane. Quantitatively (Table 3), the trained policy yields an autonomous success rate of 84%, proving that voxel-native interactive state-spaces provide effective closed-loop feedback for autonomous control tasks.

4 Conclusion

In this work, we presented Ultra GYM, a modular data-engineering framework that mitigates dynamic ultrasound data scarcity by converting static clinical 3D volumes into interactive training environments. Operating directly on clinical voxels rather than relying on cross-modality synthesis preserves native acoustic properties and eliminates domain shifts. The framework automatically synthesizes data architectures for three distinct applications: temporal sweeps for standard plane detection, exact 6-DoF spatial annotations for hardware-free 3D reconstruction, and interactive gym environments for autonomous navigation. Downstream evaluations demonstrate the platform’s utility, achieving a 79.47% overall plane detection accuracy, an 84% autonomous navigation success rate, and a competitive local pose reconstruction error of 0.12 mm. Ultimately, Ultra GYM acts as a scalable data amplifier, upcycling limited static archives to accelerate multi-task medical AI deployment.

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